

Direct Kinematics, Inverse Kinematics, and Jacobian Derivation for the Spherical Mechanism

1 Coordinate and Orientation Convention

The output of the mechanism is represented by a unit direction vector. First, the vector is computed in the internal/base frame, called frame 1:

$$\mathbf{v}_1 = Q_1 Q_3 \mathbf{v}_5 \quad (1)$$

where $\mathbf{v}_5$ is the end-effector direction expressed in frame 5.

In this mechanism, all joint twist angles are chosen as

$$\alpha_0 = \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 90^\circ. \quad (2)$$

Because of this 90° joint arrangement, the end-effector direction is not aligned with the local z_5 -axis. Instead, the end-effector direction is offset by the design angle $\beta_2 = 75^\circ$. Therefore, the vector $\mathbf{v}_5$ is defined using spherical-coordinate parameters in frame 5:

$$\mathbf{v}_5 = \begin{bmatrix} \sin \beta_2 \\ \sin \beta_1 \cos \beta_2 \\ \cos \beta_1 \cos \beta_2 \end{bmatrix}. \quad (3)$$

For the current case,

$$\beta_1 = 0, \quad \beta_2 = 75^\circ, \quad (4)$$

so

$$\mathbf{v}_5 = \begin{bmatrix} \sin 75^\circ \\ 0 \\ \cos 75^\circ \end{bmatrix}. \quad (5)$$

The final-frame direction vector is defined using the convention

$$\mathbf{d} = \begin{bmatrix} -v_1(2) \\ v_1(1) \\ v_1(3) \end{bmatrix}. \quad (6)$$

This means the final frame is related to frame 1 by

$$x_f = -y_1, \quad y_f = x_1, \quad z_f = z_1. \quad (7)$$

The final direction vector is parameterized using two orientation variables, ϕ_1 and ϕ_2 , as

$$\mathbf{d} = \begin{bmatrix} -\cos \phi_2 \sin \phi_1 \\ \cos \phi_2 \cos \phi_1 \\ \sin \phi_2 \end{bmatrix}. \quad (8)$$

Here, ϕ_1 represents the azimuth angle and ϕ_2 represents the tilt/elevation angle of the final direction vector.

Therefore, once $\mathbf{d}$ is known, the orientation angles are extracted using

$$\phi_1 = \text{atan2}(-d_x, d_y), \quad (9)$$

$$\phi_2 = \sin^{-1}(d_z). \quad (10)$$

2 Rotation Matrices

The kinematic model is built using rotation matrices. These matrices describe how the local joint frames are transformed into the base/internal frame. A picture of the studied robot mechanism is shown in Fig. 2.

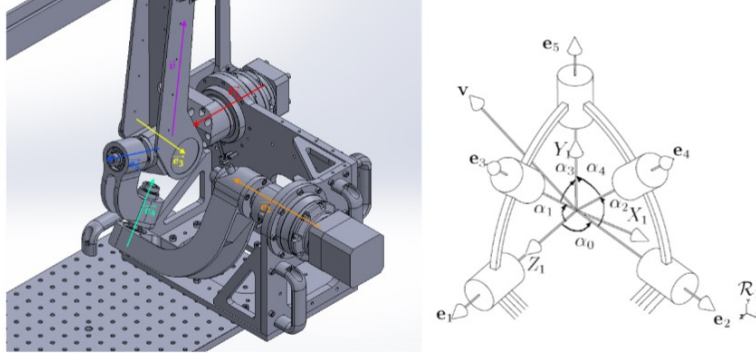


Figure 1: Spherical robot picture from old report with z-axis orientation

The constant base rotation Q_0 is defined as a rotation about the Y -axis:

$$Q_0 = R_y(\alpha_0) \quad (11)$$

with

$$R_y(\alpha) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}. \quad (12)$$

The joint rotations are represented using the DH rotation matrix:

$$Q_i = R_{\text{DH}}(\alpha_i, \theta_i) \quad (13)$$

where

$$R_{\text{DH}}(\alpha, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta \cos \alpha & \sin \theta \sin \alpha \\ \sin \theta & \cos \theta \cos \alpha & -\cos \theta \sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}. \quad (14)$$

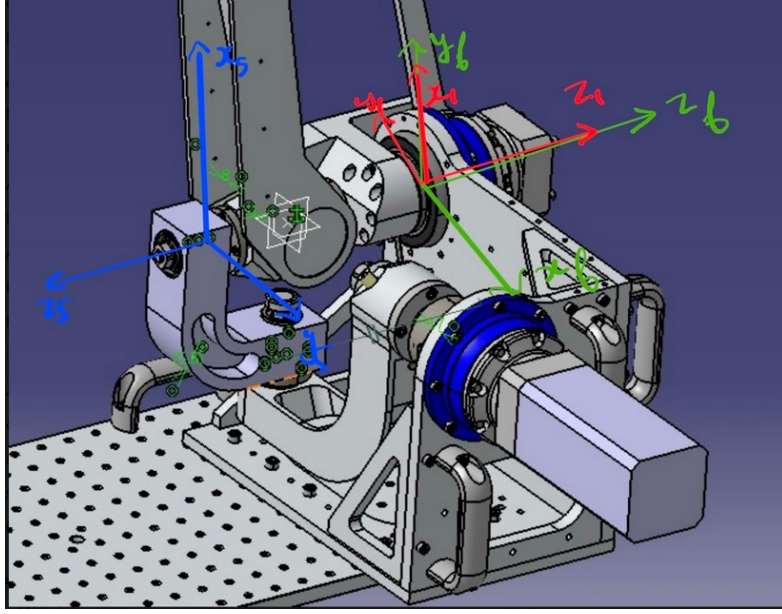


Figure 2: Spherical robot picture with frame 5, frame 1 and final frame

Therefore, for the first and second actuated joints,

$$Q_1 = R_{DH}(\alpha_1, \theta_1), \quad (15)$$

$$Q_2 = R_{DH}(\alpha_2, \theta_2). \quad (16)$$

The passive joint rotation is

$$Q_3 = R_{DH}(\alpha_3, \theta_3). \quad (17)$$

Using these matrices, the main direction vectors used in the direct kinematics are

$$\mathbf{e}_3^1 = Q_1 \mathbf{e}_{33}, \quad (18)$$

$$\mathbf{e}_4^1 = Q_0 Q_2 \mathbf{e}_{44}, \quad (19)$$

$$\mathbf{e}_5^1 = Q_1 Q_3 \mathbf{e}_{55}. \quad (20)$$

The internal output vector is

$$\mathbf{v}_1 = Q_1 Q_3 \mathbf{v}_5. \quad (21)$$

These rotation matrices are used consistently in the direct kinematics, inverse kinematics, and Jacobian derivations.

3 Direct Kinematics

The direct kinematics problem is

$$(\theta_1, \theta_2) \rightarrow \theta_3 \rightarrow \mathbf{d} \rightarrow (\phi_1, \phi_2). \quad (22)$$

The known inputs are the motor angles θ_1 and θ_2 . The unknown passive angle is θ_3 . The mechanism has two geometric constraint axes:

$$\mathbf{e}_3^1 = Q_1 \mathbf{e}_{33}, \quad (23)$$

$$\mathbf{e}_4^1 = Q_0 Q_2 \mathbf{e}_{44}, \quad (24)$$

where

$$\mathbf{e}_{33} = \mathbf{e}_{44} = \mathbf{e}_{55} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (25)$$

The vector

$$\mathbf{u} = Q_1 Q_3 \mathbf{e}_{55} \quad (26)$$

must satisfy the perpendicularity constraints

$$\mathbf{e}_3^1 \cdot \mathbf{u} = \cos \alpha_4, \quad (27)$$

$$\mathbf{e}_4^1 \cdot \mathbf{u} = \cos \alpha_4. \quad (28)$$

Since

$$\alpha_4 = \frac{\pi}{2}, \quad (29)$$

we have

$$\cos \alpha_4 = 0. \quad (30)$$

Therefore, the constraints become

$$\mathbf{e}_3^1 \cdot \mathbf{u} = 0, \quad (31)$$

$$\mathbf{e}_4^1 \cdot \mathbf{u} = 0. \quad (32)$$

This means that $\mathbf{u}$ must be perpendicular to both $\mathbf{e}_3^1$ and $\mathbf{e}_4^1$. Therefore, in the general case,

$$\mathbf{u} = \pm \frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|}. \quad (33)$$

The two signs correspond to two possible direct-kinematic branches.

3.1 Solving theta3 from u

Once $\mathbf{u}$ is known, θ_3 is computed from

$$\mathbf{u} = Q_1 Q_3 \mathbf{e}_{55}. \quad (34)$$

For the current geometry,

$$Q_1 Q_3 \mathbf{e}_{55} = \begin{bmatrix} \cos \theta_1 \sin \theta_3 \\ \sin \theta_1 \sin \theta_3 \\ -\cos \theta_3 \end{bmatrix}. \quad (35)$$

Therefore,

$$u_x = \cos \theta_1 \sin \theta_3, \quad (36)$$

$$u_y = \sin \theta_1 \sin \theta_3, \quad (37)$$

$$u_z = -\cos \theta_3. \quad (38)$$

Thus,

$$\sin \theta_3 = \frac{u_x}{\cos \theta_1}, \quad (39)$$

or, when numerically more stable,

$$\sin \theta_3 = \frac{u_y}{\sin \theta_1}. \quad (40)$$

Also,

$$\cos \theta_3 = -u_z. \quad (41)$$

Finally,

$$\theta_3 = \text{atan2}(\sin \theta_3, \cos \theta_3). \quad (42)$$

After θ_3 is known, the output vector is computed as

$$\mathbf{v}_1 = Q_1 Q_3 \mathbf{v}_5. \quad (43)$$

Then the final-frame vector is

$$\mathbf{d} = \begin{bmatrix} -v_1(2) \\ v_1(1) \\ v_1(3) \end{bmatrix}. \quad (44)$$

Finally,

$$\phi_1 = \text{atan2}(-d_x, d_y), \quad (45)$$

$$\phi_2 = \sin^{-1}(d_z). \quad (46)$$

4 Direct-Kinematic Branch Selection

Because

$$\mathbf{u} = \pm \frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|}, \quad (47)$$

there are two possible branches.

If the sign is fixed manually, for example

$$\mathbf{u} = -\frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|}, \quad (48)$$

then the DK can choose the wrong physical branch. This was the reason the DK sometimes returned results such as

$$\theta_3 \approx -135^\circ, \quad (49)$$

while IK gave the physically expected branch near

$$\theta_3 \approx 45^\circ \quad (50)$$

for the same region.

To make DK consistent with IK, both branches are computed:

$$\mathbf{u}_+ = \frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|}, \quad (51)$$

$$\mathbf{u}_- = -\frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|}. \quad (52)$$

Each one gives a possible value of θ_3 :

$$\theta_{3,+}, \quad \theta_{3,-}. \quad (53)$$

Then the branch closest to the IK/home branch is selected. The home branch is

$$\theta_{3,\text{home}} = 15^\circ. \quad (54)$$

The branch-selection rule is

$$\theta_3 = \arg \min_{\theta_{3,+}, \theta_{3,-}} |\text{wrapToPi}(\theta_3 - \theta_{3,\text{home}})|. \quad (55)$$

This makes the DK follow the same physical branch as the IK near the home pose.

5 Singularity at theta1 = 0 and theta2 = 0

At the home configuration,

$$\theta_1 = 0, \quad \theta_2 = 0, \quad (56)$$

the two constraint axes become parallel.

For this mechanism,

$$\mathbf{e}_3^1 = Q_1 \mathbf{e}_{33}. \quad (57)$$

At $\theta_1 = 0$,

$$\mathbf{e}_3^1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}. \quad (58)$$

Also,

$$\mathbf{e}_4^1 = Q_0 Q_2 \mathbf{e}_{44}. \quad (59)$$

At $\theta_2 = 0$,

$$\mathbf{e}_4^1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}. \quad (60)$$

Therefore,

$$\mathbf{e}_3^1 = \mathbf{e}_4^1, \quad (61)$$

so

$$\mathbf{e}_3^1 \parallel \mathbf{e}_4^1. \quad (62)$$

Then,

$$\mathbf{e}_3^1 \times \mathbf{e}_4^1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (63)$$

Therefore,

$$\mathbf{u} = \frac{\mathbf{e}_3^1 \times \mathbf{e}_4^1}{\|\mathbf{e}_3^1 \times \mathbf{e}_4^1\|} \quad (64)$$

is undefined.

This is the main reason for the singularity at

$$\theta_1 = 0, \quad \theta_2 = 0. \quad (65)$$

The two constraints become dependent. Instead of giving two independent equations for $\mathbf{u}$, they collapse into the same equation:

$$\mathbf{e}_3^1 \cdot \mathbf{u} = 0, \quad (66)$$

$$\mathbf{e}_4^1 \cdot \mathbf{u} = 0. \quad (67)$$

Since

$$\mathbf{e}_3^1 = \mathbf{e}_4^1, \quad (68)$$

both constraints are identical.

At home, the constraints only say

$$\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \cdot \mathbf{u} = 0, \quad (69)$$

which gives

$$u_y = 0. \quad (70)$$

This leaves infinitely many possible $\mathbf{u}$ vectors in the $x_1 z_1$ plane. Therefore, θ_3 is not uniquely determined by the DK constraints at $(0, 0)$.

This is why DK needs a home-branch rule at this point. The IK gives

$$\theta_3 = 15^\circ \quad (71)$$

at

$$\phi_1 = 0, \quad \phi_2 = 0 \quad (72)$$

because IK solves from the desired output direction. But the DK constraints alone cannot determine this value at the singular point.

Thus,

$$\boxed{\theta_1 = 0, \theta_2 = 0 \Rightarrow \mathbf{e}_3^1 \parallel \mathbf{e}_4^1 \Rightarrow \mathbf{e}_3^1 \times \mathbf{e}_4^1 = 0 \Rightarrow \mathbf{u} \text{ is not unique.}} \quad (73)$$

6 Inverse Kinematics

The inverse kinematics problem is

$$(\phi_1, \phi_2) \rightarrow \mathbf{d} \rightarrow \mathbf{v}_1 \rightarrow (\theta_1, \theta_3, \theta_2). \quad (74)$$

First, the desired final-frame direction is computed:

$$\mathbf{d}_{\text{des}} = \begin{bmatrix} -\cos \phi_2 \sin \phi_1 \\ \cos \phi_2 \cos \phi_1 \\ \sin \phi_2 \end{bmatrix}. \quad (75)$$

Then it is converted back to frame 1.

Since DK uses

$$\mathbf{d} = \begin{bmatrix} -v_1(2) \\ v_1(1) \\ v_1(3) \end{bmatrix}, \quad (76)$$

the inverse conversion is

$$\mathbf{v}_1 = \begin{bmatrix} d_y \\ -d_x \\ d_z \end{bmatrix}. \quad (77)$$

Thus,

$$\mathbf{v} = \begin{bmatrix} d_{\text{des},y} \\ -d_{\text{des},x} \\ d_{\text{des},z} \end{bmatrix}. \quad (78)$$

The IK then solves

$$\mathbf{v} = Q_1 Q_3 \mathbf{v}_5. \quad (79)$$

6.1 Solving theta1

The first equation comes from

$$\mathbf{e}_{35} \cdot \mathbf{v}_5 = \mathbf{e}_3^1 \cdot \mathbf{v}. \quad (80)$$

For this geometry,

$$\mathbf{e}_{35} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \quad (81)$$

Therefore,

$$\mathbf{e}_{35} \cdot \mathbf{v}_5 = \sin \beta_1 \cos \beta_2. \quad (82)$$

Define

$$b = \sin \beta_1 \cos \beta_2. \quad (83)$$

Also,

$$\mathbf{e}_3^1 = \begin{bmatrix} \sin \theta_1 \\ -\cos \theta_1 \\ 0 \end{bmatrix}. \quad (84)$$

Let

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}. \quad (85)$$

Then,

$$\mathbf{e}_3^1 \cdot \mathbf{v} = v_x \sin \theta_1 - v_y \cos \theta_1. \quad (86)$$

Therefore,

$$v_x \sin \theta_1 - v_y \cos \theta_1 = b. \quad (87)$$

Let

$$R = \sqrt{v_x^2 + v_y^2}, \quad (88)$$

$$\psi = \text{atan2}(v_y, v_x). \quad (89)$$

Then,

$$v_x \sin \theta_1 - v_y \cos \theta_1 = R \sin(\theta_1 - \psi). \quad (90)$$

Therefore,

$$R \sin(\theta_1 - \psi) = b. \quad (91)$$

So,

$$\theta_1 = \psi + \sin^{-1} \left(\frac{b}{R} \right), \quad (92)$$

or

$$\theta_1 = \psi + \pi - \sin^{-1} \left(\frac{b}{R} \right). \quad (93)$$

This gives two possible θ_1 solutions.

6.2 Solving theta3

Once θ_1 is known, use

$$\mathbf{v} = Q_1 Q_3 \mathbf{v}_5. \quad (94)$$

Premultiply by Q_1^T :

$$Q_1^T \mathbf{v} = Q_3 \mathbf{v}_5. \quad (95)$$

Define

$$\mathbf{A} = Q_1^T \mathbf{v}. \quad (96)$$

Let

$$\mathbf{v}_5 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad (97)$$

where

$$a = \sin \beta_2, \quad (98)$$

$$b = \sin \beta_1 \cos \beta_2, \quad (99)$$

$$c = \cos \beta_1 \cos \beta_2. \quad (100)$$

For $\alpha_3 = \pi/2$,

$$Q_3 \mathbf{v}_5 = \begin{bmatrix} a \cos \theta_3 + c \sin \theta_3 \\ a \sin \theta_3 - c \cos \theta_3 \\ b \end{bmatrix}. \quad (101)$$

So,

$$A_1 = a \cos \theta_3 + c \sin \theta_3, \quad (102)$$

$$A_2 = a \sin \theta_3 - c \cos \theta_3. \quad (103)$$

Solving for $\cos \theta_3$ and $\sin \theta_3$ gives

$$\cos \theta_3 = \frac{aA_1 - cA_2}{a^2 + c^2}, \quad (104)$$

$$\sin \theta_3 = \frac{cA_1 + aA_2}{a^2 + c^2}. \quad (105)$$

Then,

$$\theta_3 = \text{atan2}(\sin \theta_3, \cos \theta_3). \quad (106)$$

6.3 Solving theta2

After θ_1 and θ_3 are known, compute

$$\mathbf{u} = Q_1 Q_3 \mathbf{e}_{11}. \quad (107)$$

The remaining constraint is

$$\mathbf{u} \cdot \mathbf{e}_4^1 = \cos \alpha_4. \quad (108)$$

Since

$$\mathbf{e}_4^1 = Q_0 Q_2 \mathbf{e}_{44}, \quad (109)$$

and for this geometry,

$$\mathbf{e}_4^1 = \begin{bmatrix} 0 \\ -\cos \theta_2 \\ -\sin \theta_2 \end{bmatrix}, \quad (110)$$

we get

$$\mathbf{u} \cdot \mathbf{e}_4^1 = -u_y \cos \theta_2 - u_z \sin \theta_2. \quad (111)$$

Therefore,

$$-u_y \cos \theta_2 - u_z \sin \theta_2 = \cos \alpha_4. \quad (112)$$

Write this as

$$A_2 \sin \theta_2 + B_2 \cos \theta_2 = C, \quad (113)$$

where

$$A_2 = -u_z, \quad (114)$$

$$B_2 = -u_y, \quad (115)$$

$$C = \cos \alpha_4. \quad (116)$$

Let

$$R_2 = \sqrt{A_2^2 + B_2^2}, \quad (117)$$

$$\delta = \text{atan2}(B_2, A_2). \quad (118)$$

Then,

$$A_2 \sin \theta_2 + B_2 \cos \theta_2 = R_2 \sin(\theta_2 + \delta). \quad (119)$$

So,

$$R_2 \sin(\theta_2 + \delta) = C. \quad (120)$$

Therefore,

$$\theta_2 = \sin^{-1} \left(\frac{C}{R_2} \right) - \delta, \quad (121)$$

or

$$\theta_2 = \pi - \sin^{-1} \left(\frac{C}{R_2} \right) - \delta. \quad (122)$$

This gives two possible θ_2 values for each (θ_1, θ_3) pair. Therefore, IK can return multiple solutions.

7 Jacobian Derivation

The Jacobian is built from the velocity form of the constraints.

The constraints are

$$\mathbf{e}_3^1 \cdot \mathbf{v}_1 = \text{constant}, \quad (123)$$

$$\mathbf{e}_4^1 \cdot \mathbf{e}_5^1 = \text{constant}, \quad (124)$$

where

$$\mathbf{v}_1 = \mathbf{v}_{\text{internal}}, \quad (125)$$

and

$$\mathbf{e}_5^1 = Q_1 Q_3 \mathbf{e}_{55}. \quad (126)$$

All vectors in these constraints are expressed in frame 1.
The book-style velocity form is

$$K \dot{\theta} = J_0 \boldsymbol{\omega}_1. \quad (127)$$

Here,

$$K = \begin{bmatrix} k_{11} & 0 \\ 0 & k_{22} \end{bmatrix}, \quad (128)$$

with

$$k_{11} = \mathbf{v}_1 \cdot (\mathbf{e}_1^1 \times \mathbf{e}_3^1), \quad (129)$$

$$k_{22} = \mathbf{e}_5^1 \cdot (\mathbf{e}_2^1 \times \mathbf{e}_4^1). \quad (130)$$

The angular part is

$$J_0 = \begin{bmatrix} (\mathbf{e}_3^1 \times \mathbf{v}_1)^T \\ (\mathbf{e}_4^1 \times \mathbf{e}_5^1)^T \end{bmatrix}. \quad (131)$$

The task variables are ϕ_1 and ϕ_2 , so

$$\boldsymbol{\omega}_1 = S_{\omega 1} \dot{\boldsymbol{\phi}}, \quad (132)$$

where

$$\dot{\boldsymbol{\phi}} = \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix}. \quad (133)$$

Therefore,

$$K \dot{\boldsymbol{\theta}} = J_0 S_{\omega 1} \dot{\boldsymbol{\phi}}. \quad (134)$$

Define

$$J = J_0 S_{\omega 1}. \quad (135)$$

Then,

$$K \dot{\boldsymbol{\theta}} = J \dot{\boldsymbol{\phi}}. \quad (136)$$

Thus,

$$\dot{\boldsymbol{\theta}} = K^{-1} J \dot{\boldsymbol{\phi}}. \quad (137)$$

The final Jacobian mapping is

$$M = K^{-1} J. \quad (138)$$

In MATLAB, this is computed as

$$M = K \backslash J. \quad (139)$$

8 Angular Velocity Mapping

The final-frame direction convention is

$$\mathbf{d} = \begin{bmatrix} -\cos \phi_2 \sin \phi_1 \\ \cos \phi_2 \cos \phi_1 \\ \sin \phi_2 \end{bmatrix}. \quad (140)$$

This corresponds to

$$\phi_1 = \text{azimuth about } z_f, \quad (141)$$

$$\phi_2 = \text{tilt about the rotated } x_f \text{ axis.} \quad (142)$$

The angular velocity in the final frame is

$$\boldsymbol{\omega}_f = \dot{\phi}_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \dot{\phi}_2 \begin{bmatrix} \cos \phi_1 \\ \sin \phi_1 \\ 0 \end{bmatrix}. \quad (143)$$

Therefore,

$$\boldsymbol{\omega}_f = \begin{bmatrix} \cos \phi_1 \dot{\phi}_2 \\ \sin \phi_1 \dot{\phi}_2 \\ \dot{\phi}_1 \end{bmatrix}. \quad (144)$$

So,

$$\boldsymbol{\omega}_f = S_{\omega f} \dot{\boldsymbol{\phi}}, \quad (145)$$

where

$$S_{\omega f} = \begin{bmatrix} 0 & \cos \phi_1 \\ 0 & \sin \phi_1 \\ 1 & 0 \end{bmatrix}. \quad (146)$$

Because the Jacobian is written in frame 1, angular velocity must be converted to frame 1.

From the final-frame convention,

$$x_f = -y_1, \quad y_f = x_1, \quad z_f = z_1. \quad (147)$$

So,

$$\boldsymbol{\omega}_1 = \begin{bmatrix} \omega_{f,y} \\ -\omega_{f,x} \\ \omega_{f,z} \end{bmatrix}. \quad (148)$$

Therefore,

$$S_{\omega 1} = \begin{bmatrix} S_{\omega f}(2, :) \\ -S_{\omega f}(1, :) \\ S_{\omega f}(3, :) \end{bmatrix}. \quad (149)$$

Substituting $S_{\omega f}$,

$$S_{\omega_1} = \begin{bmatrix} 0 & \sin \phi_1 \\ 0 & -\cos \phi_1 \\ 1 & 0 \end{bmatrix}. \quad (150)$$

This is the matrix used in the Jacobian calculation.

9 Singularity Conditions

There are two main types of singularity.

9.1 DK Singularity

DK becomes singular when

$$\mathbf{e}_3^1 \times \mathbf{e}_4^1 = 0, \quad (151)$$

which means

$$\mathbf{e}_3^1 \parallel \mathbf{e}_4^1. \quad (152)$$

At $(0, 0)$,

$$\mathbf{e}_3^1 = \mathbf{e}_4^1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}. \quad (153)$$

Therefore, DK is singular because the constraints are no longer independent. The vector $\mathbf{u}$ is not uniquely determined.

This is the singularity caused by the constraint geometry.

9.2 Jacobian Singularity

The Jacobian velocity mapping becomes singular when K is singular:

$$\det(K) = 0. \quad (154)$$

Since

$$K = \begin{bmatrix} k_{11} & 0 \\ 0 & k_{22} \end{bmatrix}, \quad (155)$$

we have

$$\det(K) = k_{11}k_{22}. \quad (156)$$

So singularity occurs when

$$k_{11} = 0, \quad (157)$$

or

$$k_{22} = 0, \quad (158)$$

where

$$k_{11} = \mathbf{v}_1 \cdot (\mathbf{e}_1^1 \times \mathbf{e}_3^1), \quad (159)$$

$$k_{22} = \mathbf{e}_5^1 \cdot (\mathbf{e}_2^1 \times \mathbf{e}_4^1). \quad (160)$$

Even if K is not exactly singular, the mapping

$$M = K^{-1}J \quad (161)$$

can become ill-conditioned. This is checked using the smallest singular value

$$\sigma_{\min}(M). \quad (162)$$

If $\sigma_{\min}(M)$ is very small, the configuration is near singular.

10 Final Interpretation

The IK is well-defined at

$$\phi_1 = 0, \quad \phi_2 = 0, \quad (163)$$

and computes

$$\theta_1 = 0, \quad \theta_2 = 0, \quad \theta_3 = 15^\circ, \quad (164)$$

because IK starts from a desired direction and solves

$$\mathbf{v} = Q_1 Q_3 \mathbf{v}_5. \quad (165)$$

However, DK at

$$\theta_1 = 0, \quad \theta_2 = 0 \quad (166)$$

has a singularity because the two constraint axes are parallel:

$$\mathbf{e}_3^1 \parallel \mathbf{e}_4^1. \quad (167)$$

The DK constraints only say that $\mathbf{u}$ must be perpendicular to that one common axis. This leaves infinitely many possible $\mathbf{u}$ directions. Therefore, θ_3 is not uniquely determined by the DK constraints at exactly $(0, 0)$.

That is why DK needs a branch rule based on the IK-consistent home branch:

$$\theta_{3,\text{home}} = 15^\circ. \quad (168)$$

The final working approach is

Use both DK branches and choose the one closest to $\theta_3 = 15^\circ$.

(169)

This gives a DK solution consistent with the IK solution and avoids selecting the flipped branch near home.